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## **BRIDGING AI DEVELOPMENT AND COMPLIANCE: DESIGN PRINCIPLES FOR THE DOCUMENTATION OF AI**

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# BRIDGING AI DEVELOPMENT AND COMPLIANCE: DESIGN PRINCIPLES FOR THE DOCUMENTATION OF AI

*Completed Research Paper*

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## Abstract

*Artificial Intelligence (AI) has been transformative in recent decades, but its black box nature and the lack of suitable documentation guidelines make regulatory compliance difficult, thereby hindering broader adoption in practice. This paper addresses the need for effective AI documentation in organizations for auditing, transparency, and regulation adherence. Based on design science research a documentation guideline for AI applications is designed and evaluated in two use cases, involving seven expert practitioners. The resulting guideline, validated by an AI auditor, provides a comprehensive structure for documenting AI systems, emphasizing legal compliance and responsible implementation. The guideline covers AI's development process, AI's integration into business and its governance, facilitating risk detection and management of AI. The paper also introduces four design principles for AI documentation: adaptability to context, re-use of existing evidence, clarity in documentation processes, and linkage of documentation to system evidence. This research offers significant contributions to academia and practice.*

*Keywords: AI Documentation, Design Principles For Documenting AI, AI Audit.*

## 1 Introduction

Organizations and society are shifting toward digitalization, with Artificial Intelligence (AI) making strides in sectors such as medicine, banking, and auditing (Gepp et al., 2018; Hosny et al., 2018; Königstorfer and Thalmann, 2020; Krieger et al., 2021). However, AI is also presenting us with its dark sides, such as bias in grading students from comparatively poor areas (Satariano, 2020), overvaluation in real estate deals (Wille, 2021), and potential gender discrimination (Webb and Martinuzzi, 2019). The reason for these errors is, that AI applications working in real-world scenarios frequently overlook various environmental elements (Saisubramanian et al., 2022), necessitating effective management of AI's negative impacts.

To effectively manage and oversee AI systems, organizations must possess the capability to evaluate and mitigate the associated risks, as emphasized by the EU's proposed AI Act (European Commission, 2021, 2023). This process necessitates comprehensive audits of AI systems and maintaining detailed and suitable documentation to ensure transparency and accountability in AI applications (Raji et al., 2020). These reasons make the documentation of AI an essential technological prerequisite for the implementation of AI in organisations.

Especially in regulated sectors, such as medicine or banking overseeing and applying AI poses challenges, especially with IT audits for safety and reliability (Königstorfer and Thalmann, 2020). AI's complexity hampers the auditing and documentation process, often leading to a lack of proper audit certificates (Arrieta et al., 2020). As governments worldwide, including the EU with its AI Act (European Parliament, 2024) and the US with new regulations and bias standards (NIST, 2022; President J. Biden, 2023), introduce AI regulations, even less regulated fields will face these issues. The challenges in AI auditing and documentation, combined with the advent of strict regulations, are increasing uncertainties, thereby hindering AI adoption.

In addition to the auditing, suitable documentation is also needed for maintaining the AI system and for training of employees. In this regard, research has focused on explainable AI (Gashi et al., 2022) and dependable process models (Wolf, 2020) to make AI more accessible. . However, in addition to making the models more transparent, a documentation of the entire development process for audit evidence is needed, but underexplored. Traditional software documentation fails to provide suitable evidence for AI audits (Appelbaum et al., 2017), with existing AI documentation proposals being scattered and non-standardized (Königstorfer and Thalmann, 2021). Thus, there is a need for transparent documentation processes and guidelines, and this paper addresses this research question:

*What should design principles for the documentation of AI look like?*

To answer this question, the design science research (DSR) methodology (Hevner, 2007; Peffers et al., 2007) was followed. After having determined the requirements for AI documentation from literature, auditors, and domain- and IT experts, this study delves into the in-depth documentation of AI in two use cases, where IT experts document an AI system, and auditors evaluate the system based on the provided documentation. The case study also develops four design principles for the documentation of AI.

## **2 Background**

AI is a key driver of digital transformation in highly regulated sectors such as finance and healthcare (Badrinath et al., 2016; Bhatia et al., 2020). AI robo-advisers in finance provide automated investment services (Bhatia et al., 2020), while in healthcare, AI assists with diagnoses and drug development (Badrinath et al., 2016; Di He et al., 2022), impacting hospital practices and healthcare roles. However, AI can also adopt historical biases (Barocas and Selbst, 2016) and, disturbingly, even peer-reviewed AI models may contain significant errors (Gundersen and Kjensmo, 2018; Kapoor and Narayanan, 2022; Roberts et al., 2021), affecting their reliability and application. Mistakes in AI development and biases in AI systems often arise from inadequate testing, irrelevant attributes, and duplicated data (Arp et al., 2020; Roberts et al., 2021). Still, these errors often remain undetected, since crucial elements of the study's framework, such as the data involved, and the design decisions, often remain either inadequately explained or inaccessible (Gundersen and Kjensmo, 2018). Proper documentation of AI applications could help in identifying and correcting such errors.

A good documentation is also important for other reasons. Employers want to document their AI applications well, because a good documentation increases the transparency of the development process and can help new employees learn how an AI was developed, how the AI can and should be used (Königstorfer and Thalmann, 2022), making it less likely that the organization suffers from employees leaving the company. A suitable documentation is also essential to implement, manage and oversee AI systems in organisations as well as to evaluate and mitigate the associated risks, as emphasized by the EU's proposed AI Act (European Parliament, 2024), and the USA's AI related standards and regulations (NIST, 2022; President J. Biden, 2023). In combination with the stark differences between AI and traditional software, the legal requirements on AI restrict AI's adoption in practice, especially in regulated sectors such as medicine, banking, and auditing due to documentation- and auditing challenges (Königstorfer and Thalmann, 2020, 2022; Krieger et al., 2021). A suitable documentation of AI applications also enables the verification of compliance with ethical guidelines (Mohammad, 2021) and supplies vital information to auditors, who can use the documentation to find errors. AI documentation is especially important in regulated scenarios where external audits are essential (Ellul et al., 2021; IEEE, 2008; Laporte and April, 2018; Marten et al., 2020; Raji et al., 2020). External audits verify IT systems'

compliance with laws (IEEE, 2008). Although AI audit guidelines exist, challenges persist. Firstly, legal compliance is not always addressed in current guidelines. For instance, the framework by Raji et al. (2020) emphasizes fairness, accountability, and transparency but neglects legal aspects. Secondly, external auditors often get less information than internal ones (Raji et al., 2020). AI documentation helps address these challenges by providing insights into (1) how the AI was developed, and (2) how the AI can and should be integrated into business processes. Yet, the lack of documentation guidelines hinders AI audits and, in turn, slows adoption in regulated industries.

However, documenting AI differs from traditional software. Software engineers view documentation as “*creation and keeping current of an artifact holding information about architectural decisions for stakeholders in an accessible, nonambiguous form*” (Clements et al., 2011). However, although traditional software engineers already have documentation standards, AI lacks suitable documentation standards, and the differences between AI and traditional systems make guidelines for traditional software engineering less meaningful (Königstorfer and Thalmann, 2021). First, AI models establish their decision making processes from the training data (Gebru et al., 2021; Huang et al., 2004; Mitchell et al., 2019), reducing the need for programmer inputs (Huang et al., 2004). This trait makes traditional documentation such as annotated source code less effective for AI (Garousi et al., 2015). Second, AI’s autonomous decision-making from data presents challenges in explaining and documenting its reasoning (Gashi et al., 2022), a problem known as the “Black Box” dilemma (Vuković and Thalmann, 2022). The intricacy of AI models, such as neural networks, complicates understanding their internal logic (Castelvecchi, 2016; Gashi et al., 2022). Consequently, when subjected to unfamiliar scenarios or data, AI systems might demonstrate unanticipated or novel responses. AI’s unpredictability in new situations creates documentation challenges, as these behaviours are hard to predict and record in advance, deviating from traditional software methods and hindering AI’s practical application (Kloker et al., 2022; Königstorfer and Thalmann, 2021; Kvamme et al., 2018). For example, some generative AI models, such as Large Language Models (LLMs), have been observed to generate hallucinated or inaccurate responses (Shi et al., 2023; Zhang et al., 2023).

From the perspective of regulators, experts from practice and scholars, documentation must meet minimum requirements agreed upon by researchers, practitioners, and regulators. These encompass a general description of the AI’s application domain, documentation of training data, and information on design and development (European Parliament, 2024; Gebru et al., 2021; Mitchell et al., 2019). Additional requirements include understandability for third parties and balancing the benefits and efforts needed for documentation (Königstorfer and Thalmann, 2021). However, even though the requirements for the AI documentations are known (European Parliament, 2024; Königstorfer and Thalmann, 2021), the question of how to fulfil all of them remains. Reviewing the existing documentation guidelines does not help to illuminate the entire subject. On the one hand, detailed documentation guidelines focussing on individual aspects of the AI development process such as either the data (Gebru et al., 2021) or the model development process (Alberti et al., 2019; Beg et al., 2021; Mitchell et al., 2019; Mora-Cantalops et al., 2021) and documentation guidelines for a small handful of industries (Artrith et al., 2021; Liu et al., 2020; Rivera et al., 2020) exist. On the other hand, research has shown that AI documentations need to contain information on the entire AI development process and that governance and security aspects surrounding the AI need to be documented as well (Königstorfer and Thalmann, 2021). Existing documentation guidelines contain limited information on these aspects. Also, while the general documentation guidelines may be sufficient to get AI applications adopted in some use cases, the lack of industry specific documentation guidelines causes companies to not adopt AI (Königstorfer and Thalmann, 2020). Yet, the existing very industry specific guidelines do not or generalize well (Liu et al., 2020; Rivera et al., 2020). Hence, it is necessary to investigate how suitable guideline for AI documentation need to look like. The challenge of documenting AI well enough for regulators and auditors in combination with the requirements for AI documentations represents the problem space that this study is attempting to address.

### 3 Methodology

In this study, we develop and present two design artifacts. Our first design artifact, namely the AI documentation guideline, provides AI developers with information on how to suitably document AI applications in a way that is deemed suitable by an AI auditor. Our second design artifact, namely the (four) design principles for the documentation of AI applications informs AI developers on how to use and adapt the proposed documentation guideline with the goal of providing suitable evidence for auditors. The guideline and design principles address documentation needs for AI apps (Königstorfer and Thalmann, 2020, 2022), aiming to guide developers in documenting AI adequately for auditors' requirements. It also aligns with expert practitioners' demands (Königstorfer and Thalmann, 2021), defining the problem space according to vom Brocke et al. (2020).

To develop the documentation guideline and the design principles, this paper follows the DSR methodology (Hevner, 2007; Peffers et al., 2007) and aims to propose actionable guidance on the documentation of AI. Figure 1 presents the methodology of this study. The challenge that triggered this study was the lack of suitable documentation guidelines for AI applications to provide suitable input for audits. As McCarthy et al. (2020) state, the successful identification of problems for practitioners in DSR requires the involvement of experts in the research project. This represents the first step in Figure 1. In step two, the objectives and requirements for the documentation of AI need to be identified. The challenge of documenting AI applications as well as the requirements for AI documentations were identified using not only the literature, but also a set of empirical studies (Königstorfer and Thalmann, 2020, 2021, 2022). The requirements were used as input for the design of the documentation guideline.

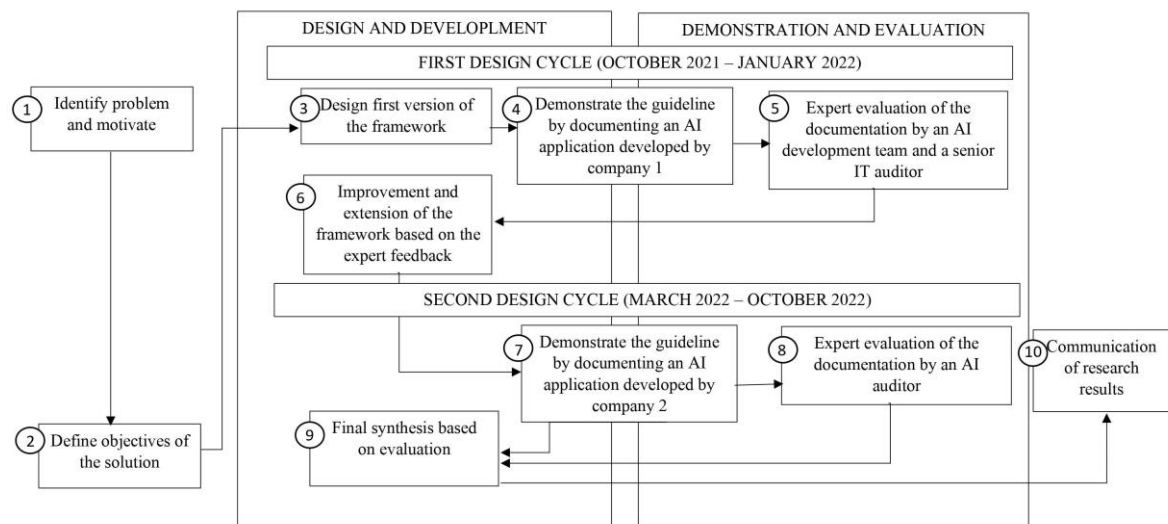


Figure 1. Summary of the procedure

The first design cycle was conducted together with experts from company one. Company one is a large, international audit company. The experts from the audit company were selected for the study due to their knowledge of the relevant legal environment as well as their experience with IT auditing. One financial auditor, one senior IT auditor and the experts from the company one's data science team were involved in this design cycle. Information on the experts involved in the study is available in Table 1.

| Name     | Job title                   | Years of experience | Employer  |
|----------|-----------------------------|---------------------|-----------|
| Expert 1 | Data Scientist / AI Auditor | 5-10 years          | Company 1 |

|             |                                                                             |                               |           |
|-------------|-----------------------------------------------------------------------------|-------------------------------|-----------|
| Expert 2    | Senior IT auditor                                                           | >20 years                     | Company 1 |
| Expert 3    | AI compliance officer                                                       | <5 years                      | Company 2 |
| Expert team | product manager, a statistician, a solution architect and a UX/UI designer. | 2 * <5years and 2 * 5-10years | Company 1 |

Table 1. description of the experts

In the third step of the study, a draft of the documentation was designed together with company one. The draft of the guideline was based on the existing knowledge space, consisting of (1) the requirements for AI documentation and (2) existing methods for documenting AI applications. The existing documentation guidelines were collected during a rigorous review of the existing literature on the topic of AI documentation and represent the solution space at the beginning of the study. Furthermore, efforts were made to search for design principles pertaining to AI documentation within academic literature, aiming to diversify the solution space. Regrettably, none were uncovered. The initial sections of the guideline were chosen based on a set of requirements for documenting AI (Königstorfer and Thalmann, 2021). Questions relating to the training data and the design decisions were based on documentation guidelines specifically made for documenting data and AI models (Arnold et al., 2019; Bender and Friedman, 2018; Gebru et al., 2021; Holland et al., 2018; Miceli et al., 2021; Mitchell et al., 2019) as well as data-type and domain-specific documentation guidelines (Artrith et al., 2021; Liu et al., 2020; Norgeot et al., 2020; Rivera et al., 2020; Walsh et al., 2021). For instance, Gebru et al. (2021) include questions on, among other things, the purpose for which the dataset was initially created and potential errors in the dataset in their documentation guideline. As a result, questions on these aspects were also included in the first draft of the documentation guideline. The questions on the application domain of the AI application were based on documentation guidelines from medicine and chemistry (Artrith et al., 2021; Liu et al., 2020; Norgeot et al., 2020; Rivera et al., 2020; Walsh et al., 2021), documentation guidelines for reviewing AI applications for compliance with ethical standards (Mohammad, 2021), and internal documentation guidelines from company 1. For instance, Liu et al. (2020) ask developers to explain and document how the AI's predictions will be integrated into the decision-making process and whether a human will validate the prediction before any action is taken. Hence these questions were included in the first draft of the guideline.

In the study's fourth step, the guideline was applied to document one AI application in company one. The AI application is aiding the company's auditors in the so-called test of details. The documentation was crafted by an AI developer and AI auditor from company one (Expert 1). The provided feedback gave insights into the developer's viewpoint. Once the documentation was complete, it was showcased to and assessed by the AI development team of six (Expert team) and a senior IT auditor (Expert 2), marking the fifth step. The IT auditor's review offered crucial understanding into the documentation's advantages, challenges, and aptness to furnish audit evidence. The senior IT auditor also provided us with crucial feedback on how the guideline can be improved. Expert 2 noticed that some of the questions in the initial draft of the guideline should be moved to a new section on AI governance. For example, the question "Were the users trained in the use of the solution?" needed to be moved to the new section on AI governance. Additionally, Expert 2 pointed out that several of the questions that were taken from the existing literature were too long, too vague, or not sufficiently informative to be useful in an audit. For instance, regarding the question "Were the users [of the AI solution] trained in the use of the solution?" the senior IT auditor pointed out that it is also important to ask, if the training program the employees participated in was certified as well. The guideline was then modified in the sixth step based on feedback from Expert 2 and evaluation from Expert 1. In response to the feedback mentioned above, for instance, several questions were relocated to a newly created category called "AI Governance", and the question "Was the training program certified by internal or external organizations or organizational units?" was added to the guideline.

In a second design cycle, the improved documentation guideline was then used to document an AI application created by company two. Company two is an AI development company that created an AI application for a company producing medical/pharmaceutical products. The AI application assists

factory line workers in quality management tasks. The documentation of the AI application was created by an AI expert of company two (Expert 3), representing the seventh step in the study. The expert from the AI development company was selected due to his knowledge and experience in developing AI applications. The use case was selected by the expert from company two. Again, basic information on the expert involved in the study is available in Table 1. Expert 3 documented the AI model using inputs from his coworkers, representing the 7<sup>th</sup> step in Figure 1.

During the documentation creation, Expert 3 also provided a list of comments on the documentation process and recommendations for improvements to the guideline. For instance, the question “Which audit evidence/business insights are being generated?” was too specific to the audit case from the first design cycle and did not make sense in the pharmaceutical context. In the eighth step, the AI application documentation was reviewed by an AI auditor of company one (Expert 1). During step 8, the AI auditor (Expert 1) found that the documentation from company 2 is of a sufficiently high quality to “*understand how the system works and whether it was designed with enough expertise*” (Expert 1), which is why we have not added another evaluation step or integrated an additional case into this study. In the ninth step, the guideline was finalized, and small improvements were made to the guideline. Step 10 in Figure 1 involves publishing and thereby communicating the results of this case study.

During the study, multiple data types were collected. Notes from guideline development included comments on existing guidelines, background on the initial design cycle use case, and company one’s processes. Feedback sessions notes, comments on company two’s documentation, recommendations, an interview, and emails with Expert 3 were gathered. In step 8, documentation evaluation, feedback, and recommendations were collected. Sensitive information was redacted. All data (interviews, notes, documentation) were in English or German, with quotes translated to English post-analysis.

To identify the design principles, an informed inductive coding approach per Mayring (2021) was used. Initial codes stemmed from the interview studies (Königstorfer and Thalmann, 2021, 2022) and literature review. For instance, practitioners find rigid documentation guidelines hinder innovation, leading to the “flexibility of documentation for ensuring innovation” code. New codes arose during analysis, such as “incorporation of use case-specific questions” due to varied regulatory requirements and AI integration descriptions. Codes were ultimately merged; e.g., “incorporation of use case-specific questions” combined with “flexibility of documentation for ensuring innovation” to form “flexibility to adapt to specific contexts”. We use Accountability Theory (Vance et al., 2015) as our kernel theory.

## 4 Results

The results are split into two subsections. The first subsection presents the documentation guideline and insights from the two design cycles. The second subsection presents the design principles that were extracted from the data.

### 4.1 Documentation guideline

The final guideline consists of (1) a cover sheet, (2) the guideline, and (3) any additional applicable and required (regulatory) documents.

The **cover sheet** should enable auditors (A) to get a suitable overview of the rules and regulations related to the use case and the AI application, and (B) get a basic overview over the AI application and its associated risks. The cover sheet contains in total 19 questions on basic information of the AI, such as the AI application itself, the use case in which the AI is applied, the data that the AI is trained with and exposed to after its deployment.

The **guideline** encompasses 106 questions across four distinct aspects of the AI development process. In its current form, the guideline is domain independent and covers all aspects of the AI development process, the business- and IT environment that needs to be documented for an audit. The names of the sections as well as a description of these sections can be found in Table 2.

| Section | Description |
|---------|-------------|
|---------|-------------|

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Application Domain | This subsection contains 24 questions related to the business understanding and IT environment forming the basis of the AI application. Specifically, the questions relate to the following four aspects: an introduction of general use case and integration of the AI to the business process and software; legal aspects of the AI application within the specific use case; evaluation of AI performance in the business context / application case; and information on the hardware and interaction with other software components during training and deployment. |
| Data               | The "Data" section provides auditors with insights into assessing the quality and legality of all data aspects used by the AI application. The 46 questions in this section relate to all aspects of the dataset that is used to train the AI model, including data collection process(es), data cleaning procedures; data/feature preparation; data quality; data sharing; data maintenance; and legal aspects (i.e. GDPR compliance during data collection) related to the dataset.                                                                                   |
| Design Decisions   | The "Design Decisions" section provides auditors with insights into the technical decisions made during the AI development. The questions in this section relate to the model training (i.e. choice of AI model, including final hyperparameters); intended use and evaluation tests of model; as well as legal aspects (i.e. bias tests ; availability of an explanation as specified in the GDPR).                                                                                                                                                                    |
| AI Governance      | The nine questions in this section cover aspects related to the governance of the AI application, including prevention and consequences of misuse, oversight mechanisms, and ethical compliance.                                                                                                                                                                                                                                                                                                                                                                        |

Table 2. summary of guideline

Regarding the content of the guideline, the experts found the guidelines to be well-structured and comprehensive, covering all critical aspects of AI implementation as one expert explained:

(Question: Does the guideline make sense?) A: *“Definitely yes. Just going by this structure of the guidelines... That you start with the application domain, how it looks, what [the AI application] is supposed to do, how it is integrated into the software. Then, of course, there's the big topic of getting data, how is it collected, how is it processed. Later on, it is stated how the actual logic in the model works. [...] What I also found great is this aspect of governance that was included. So there was also a very comprehensive impression of what the AI should ultimately ensure”* (Expert 3)

The expert highlights that not just technical aspects are important but also managerial aspects such as the AI governance and that the guideline can be used with ease by AI developers. Additionally, it was mentioned that working with the guideline has the positive side effect that developers thinking more about the potential downstream impact of the AI application from a business perspective.

Regarding the applicability of the guideline in audits, experts highlighted legal and compliance issues related to the documentation of AI. This was especially the case if the AI was based on critical knowledge as one interviewee explains:

*“You will often have to deal with strict confidentiality agreements.[...] I would also take this into account in the questions and keep in mind that answers to some questions cannot be released”* (Expert 3)

This expert highlights that the documentation could contain business critical knowledge and thus be classified. As a result, NDAs may restrict the amount of information that can be put into the documentation or with who the documentation can be shared. Also the willingness to document the AI appropriately seems limited if critical knowledge should be shared. However, even if the answers to some questions are unavailable, the auditor can deem the documentation suitable for an audit.

*“I found the answers from [company 2] still rather superficial, but in the end, it was enough for me as an expert to basically understand how the system works and whether it was designed with enough expertise.”* (Expert 1)

This shows that a documentation that was created using the guideline is suitable according to the standards of an AI auditor. Expert 1 highlights the challenging trade-off auditors have to make and thus



also the decision on which level of detail (superficial vs. detailed) the guidelines will be. As describing more vague dimensions, such as the motivation from a business case or business impact always contains some room for interpretation, this question on the level of detail needs to be clarified with the auditor.

The insight from the two design cycles shows that the process of documenting and auditing AI remains complex and that not everything can be formally specified, but the final documentation guideline represents an important step towards addressing the challenge of documenting and auditing AI.

## 4.2 Design Principles

Based on the insight from the interaction with experts involved in the development and documentation of the AI, the following design principles are elicited. All involved parties highlighted the challenge of designing meaningful documentation for the audit without creating huge burden on all involved parties.

### 4.2.1 Design Principle 1 – Flexibility to adapt the guideline to specific contexts

In the process of documenting the AI applications, practitioners repeatedly pointed to different questions in the documentation guideline that are only relevant under specific circumstances. Using personal data to train the AI, for instance, requires AI developers to document information on how the requirements stated in the EU's General Data Protection Regulation (GDPR) are fulfilled. This includes information on how the decision of the AI model is explained, a significant challenge given the Black Box character of AI. If no personal data is used, GDPR related questions do not need to be addressed. Similarly, AI applications in the cloud need to be documented differently than AI applications deployed on servers located on the premises of the company. Since company one is using the cloud to develop their AI, Expert 2 points to the need to integrate the provider's documentation, thereby adjusting the documentation to the company's use case:

*„At this point, [we may need to] refer to the certification of cloud platforms that was used (e.g., [our] Azure instance), e.g., via the Service Organization Control (SOC) Report“ (Expert 2)*

Similarly, all involved experts stated that domain- and application-specific requirements are very important. Thus, apart from general and domain-independent elements in the documentation guideline, the need to add questions related to domain-specific regulatory issues or business requirements became apparent. In both design cycles the involved practitioners highlighted the need to integrate questions to comply with domain-specific laws and regulations as well as industry standards. In case 1, the inclusion of several questions in the first draft of the guideline to a documentation guideline for traditional automated tools and techniques (ATT) in audit was necessary for example. This documentation guideline for ATTs specifically asks creators of traditional software to keep track of certain aspects related to the software, such as the risks the AI application was supposed to analyze.

During steps 3 and 4 in Figure 1, Expert 1 emphasized the importance of integrating a set of audit specific questions in the documentation to satisfy audit specific standards and regulations as well as questions from the internal certification department. Yet, during steps 7 and 8 in Figure 1, Expert 3 pointed out that many of the audit specific questions made little sense for his employer. Expert 3 also argued that since his employer created AI applications for its clients, different information about the AI applications needed to be included in the documentation to account for the specificities of the client.

*“I am missing the following topics in the guideline: [...] Integration of the AI application at the customer, i.e. how is the application used (e.g. internally or externally), what do interfaces look like, etc.” (Expert 3)*

*“It would make a difference whether an AI application meets specific requirements. Such as: Black and White Box, Cloud-based SaaS or on-premises solution, or degree of autonomy of the AI application If so, this would also need to be considered in/added to the documentation. ” (Expert 3)*

Expert 3 highlights, that always some flexibility of the guidelines is needed. It became clear, that every domain has its own documentation needs, and it is challenging to cover these requirements domain-

independently with general dimensions. Based on this insight, the following design requirement was elicited:

*The documentation guideline should be flexible enough to extend the documentation to new contexts, by adding use case specific questions.*

The developers note that this adaptability is required to ensure that all relevant aspects of the AI application can be well documented.

#### 4.2.2 Design Principle 2 – Re-use of existing evidence

The case studies revealed the significant effort required for AI documentation and auditing, emphasizing the need to reduce redundancies and reuse resources. The complex development process of AI often involves various third-party data sources, cloud computing systems, and external, pre-trained foundational AI models. For these resources, documentation and certifications may already exist, which makes reusing existing audit certificates and documentation crucial for verifying compliance, completing development efficiently, and passing audits. Expert 2 explained that the integration of audit certifications of cloud solutions needs to be integrated into AI documentations to pass an audit. Reusing existing documentation is essential for many development teams to complete the development process on time. In this regard, one auditor stated:

*“I don't think a process or system would need to be audited again if another independent audit already exists. Ideally, you leverage that for your own purposes. Perhaps it depends a bit on who did the previous audit and whether one can (or wants to) rely on the fact that the work was done cleanly.”*

(Expert 1)

What the auditor also highlighted is that an informal description of the past audit is vague and may need to be reevaluated. Thus, a formal, accountable and approvable integration of audit certificates would be desirable. This leads to the following design principle:

*It needs to be possible to integrate existing evidence, such as audit certificates or documentation of AI components, into the documentation.*

The design principle highlights that documentation procedures should be able to incorporate existing audit certificates and documentations, when available. Again, abiding by this principle can save the auditor as well as the developer a significant amount of time and money during the documentation and the audit of the AI application.

#### 4.2.3 Design Principle 3 – Clear documentation processes

Especially from the developer's perspective, the challenge of having clear documentation processes was highlighted. AI applications are often created in interdisciplinary teams, with people from different domains involved in the AI development process. The challenge of this interdisciplinary collaboration is, that the knowledge is distributed, and nobody knows every detail about the AI's development, its behaviour, and the business implications. However, AI specificities make this issue particularly challenging. For instance, if hidden biases in the training data are not documented and communicated to the developers working on the data preparation and model creation steps, these biases may creep into the final model and may cause significant damage to the users. Hence, a clear description of the documentation needs and documentation processes itself is needed. This is especially relevant as related information is collected from multiple different stakeholders as Expert 3 stated:

*“You have to collect the information in different departments. First, a project manager himself has to bring his aspects to the table. Second, the machine learning engineer and all other stakeholders have to bring the rest and document the AI application”* (Expert 3)

The AI expert highlights the complexity of the documentation and points to the fact that this is not just a task for AI developers. Collecting this information from all those sources is challenging, since a lot of communication is required to create the documentation and to keep it up to date.

Additionally, the documentation is often written after the start of the project. This resulted in the employees often having to create parts of the documentation long after the decisions were made and when the reasons for making certain decisions were not fresh in the decision maker's minds anymore as one expert highlights:

*“And with our team, there is the additional challenge that the documentation is written after the project has started. Some parts we already had, others we had to rework. So that's certainly a challenge when you do it afterwards.”* (Expert 3)

The AI expert refers to a common practice in software development in general and demanded clear documentation tasks and processes helping to avoid such ex-post documentation. However, this practice made documenting the AI not just challenging but also less reliable. However, collecting the information during the development process is considered a significant challenge for AI developers as the development flow might be interrupted. Thus, finding a balance is needed and based on the insights, the following design principle is formulated:

*Clear documentation processes should specify the responsibilities as well as the point in time for documentation in the AI development process. The documentation should be made available to the relevant stakeholders in other departments promptly.*

#### **4.2.4 Design Principle 4 – The documentation needs to be linked to system evidence**

From the audit perspective, the documentation is an essential first step during the audit of AI applications. However, the documentation is not the only input needed for an audit. Auditors need to verify that the system does what the developers state, which requires them to evaluate additional evidence that can help them verify that the AI was created in the way described in the documentation and that the AI works in the desired way. The information from the documentation can be verified with the help of specific evidence from the development of the system and the environment into which the AI is integrated. Given the specifics of AI, the focus of auditors regarding the system evidence is on aspects such as the data preparation process, evidence from the training process of the AI, instructions from the developers, and other issues related to the development process. Expert 1 describes the situation in the following way:

*“If I really tried to attest a system, I would have to be able to verify the information from the documentation somehow; e.g. by a reconstruction of the model training (this of course also has its limitations, if one thinks e.g. of large language models), a tracing of the data flow within the system (data lineage / audit trail), or an inspection of the source code.”* (Expert 1)

This implies two things: Firstly, the documentation should reflect the latest state of the system. AI solutions are often changed or updated after deployment. These changes need to be included in the documentation in a timely manner. As a result, methods are needed to keep the documentation updated and aligned with the current implementation of the AI solution. Secondly, the auditor needs to be able to connect the statements in the documentation – wherever possible – so that evidence can support the documentation, e.g., to training logs or intermediate data outputs. Hence, the following design principle emerged:

*The documentation guideline should consider the integration of supportive evidence to cope with the evolution of the AI application.*

Linking the documentation directly to the system evidence allows the auditor to verify the information in an efficient and effective manner. It also seems to be a promising way to audit AI applications in an efficient way and may also save the developer time.

## 5 Discussion

The paper argues that the documentation of AI is key for assessing the impact of AI on organizations during audits and also on society in a broader sense. For this purpose, this paper has two key contributions: (1) a documentation guideline, and (2) four design principles for documenting AI applications. As research has shown, the documentation guideline represents a much-needed, industry neutral standard for AI developers and auditors, covering not only the AI development process but also the integration of AI applications into business processes and the safety and governance process surrounding the AI. The paper's perspective on AI governance, which has so far received limited attention in the literature on documenting AI, is a new perspective. The guideline has been validated and found suitable for an audit by an AI auditor, making it a useful approach for practice and a first step towards addressing forthcoming regulatory needs, such as the EU AI Act. As a result, the guideline enables auditors and AI developers to detect and to manage potential risks associated with the AI system and to prevent harm before it happens. The validated guideline has a high practical value for AI developers. First, it allows developers to focus their documentation efforts on aspects that actually matter to regulators and editors. As a result, the development process becomes more efficient. Second, the validation of the guideline by an AI auditor reduces the legal uncertainty around how the requirements from the AI Act can be met. This is highly beneficial, since explicit guidance on how to meet the regulatory requirements is still outstanding.

The guideline is an important design artifact which can provide guidance for documentation of AI in organizations. As highlighted in the findings, it may be necessary to incorporate additional questions specific to individual use cases into the guideline to accommodate specific domain characteristics or scenarios encountered during the development phase. This is also supported by existing literature and regulation. For instance, literature suggests that transformations in information systems' infrastructure, including the adoption of cloud and distributed computing, along with evolutions in AI development processes, such as regular or almost continuous retraining with new data, may result in ML deployments that contradict foundational theoretical assumptions of ML algorithms (Parker, 2012). Additionally, researchers point out that specific aspects of the AI development process may be relatively more important in one field than in others (Rostamzadeh et al., 2022). Though domain-specific regulatory frameworks are argued for (Ellul et al., 2021), the importance of domain-specific adjustments for AI documentation and auditing remains understated. The guideline does not encompass every conceivable question that an auditor may need, necessitating additional modifications. However, the documentation guideline does provide researchers and experts from practice with a solid industry neutral foundation, to which additional use case specific questions and answers can be added. Adapting guidelines for non-regulated use cases requires further research on how to create use case-specific documentation questions.

Incorporating existing evidence in documentation is also seen as essential by other researchers and experts from practice, for instance in AI documentation guidelines for medical practitioners (Liu et al., 2020; Rivera et al., 2020) and audit reports required by existing standards (ISO and IEEE, 2014). However, re-using documentations and audit certificates in AI applications presents difficulties regarding the reuse of documentation and audit certificates for AI based transfer learning and subcomponents of AI systems. However, repurposing existing documentation is challenging as fine-tuning alters the model, potentially leading to different behavior. This raises questions about the relevance of existing documentation and certifications for the new version, and strategies for integrating it with new training data. Also, the question of how to suitably audit and certify the subcomponents of AI applications or foundational AI models is still open. Exploring the integration of audit certificates into AI documentation is a promising avenue for future research.

The design principles represent the second major contribution to the literature. To the best of the authors' knowledge, the paper is the first to discuss design principles for the documentation of AI. The design principle of linking documentation with system evidence contributes to the literature on auditing AI applications. Linking documentation to system evidence such as source code may address challenges such as unclear audit evidence collection. For instance, Appelbaum et al. (2017) point to missing information related to the collection and the information content of suitable audit evidence, which could

be avoided with a suitable documentation. A comprehensive documentation with explicit system evidence links aids auditors in their analysis. However, situations where the developer lacks full access to the training resources or the system on which the AI is deployed or cannot share information due to NDAs or intellectual property rules pose challenges. Further research is needed to explore solutions for these scenarios.

The design principles also link well to research on documenting AI. Königstorfer and Thalmann (2021) point out that a suitable tradeoff between benefits and the efforts going into the documentation is challenging and needed for companies to operate efficiently. Design principles two and three are a significant step into that direction. Clear documentation processes and the re-use of existing evidence can cut down on the time needed to create the documentation, while maintaining a high degree of documentation quality for the auditors.

The importance of well-defined documentation processes has also been underscored within the realm of software engineering. In this domain, it has been evidenced that implementing a structured approach to technical documentation can enhance a company's quality assurance efforts (Mashburn and Vantrease, 1983), while proficient documentation can also aid in adhering to legal requirements (Tran et al., 2012). However, despite the existing discourse on documentation processes and their formalization within traditional software engineering literature, as exemplified by studies such as (Valtanen et al., 2009; Visconti and Cook, 2004), this aspect has not yet been addressed within the context of AI to the best of our knowledge. A well-structured formalization of the documentation process is imperative for effectively documenting AI systems. Further investigation is warranted to delve into the formalization of documentation processes specific to AI.

The design principle of linking the documentation to the system evidence can help protect users and society from harm caused by AI applications that were built incorrectly. The design principle can also assist developers in swiftly rectifying their errors. Studies indicate that numerous academic papers developing AI contain mistakes, often relating to fundamental aspects of AI development such as the omission of a separate testing dataset critical for model validation (Arp et al., 2020) and the use of invalid or unsuitable attributes, which act as substitutes for the target variable, during AI model creation (Arp et al., 2020). This connection between documentation and evidence can enable quicker detection of errors in the development process and correction of these errors by AI researchers and developers, thus providing a clear benefit to not only the research community, but also to society as a whole.

Research should also examine how generative AI and automatic documentation tools such as the one included in Microsoft Azure (Microsoft, 2022) can complement human efforts. Previous research has shown that automatic documentation can potentially save developers a significant amount of time (Wang et al., 2021). However, some generative AI models such as LLMs are known to hallucinate and to provide made up or incorrect answers (Shi et al., 2023; Zhang et al., 2023), making the answers potentially unsuitable for audits. Further studies are required to optimize documentation processes involving human and computer-based documenters.

This paper's limitations include only two design cycles and not covering companies using continuously learning AI models. Also, quotes were translated from German to English. However, two authors double checked for an appropriate translation. The study's scope is limited to Germany and Austria, and other countries may face different regulatory requirements. At the time of this paper, the EU's AI Act and other AI regulations are still work in progress. Thus this work could be used as input for the implementation process, but future research should also investigate if future adjustments to the documentation guideline are needed.

## **6 Conclusion**

This study sets out to answer the question of what design principles for the documentation of AI need to look like. The primary contribution of the paper consists of the four design requirements that emerged from two cases and a design artifact in the form of a documentation guideline. The second contribution was a documentation guideline, which was used to document two different AI applications. Finally, first

evidence that the documentation created in the two cases is suitable for auditing these two AI applications was collected.

This study has high practical value as it can help companies detect AI related risks as well as achieve regulatory compliance. The results inform AI development teams on how to create their documentation and to design their documentation process. Auditors benefit from documentation that helps them evaluate the skill and the knowledge with which the AI has been designed and developed as well as from a suitable link to the evidence they need to evaluate the system. As a result, the study brings practitioners one step closer to meeting the upcoming regulatory requirements for AI applications and represents a significant contribution towards the technological prerequisites for implementing AI in organizations.

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